👑 CODERKING STUDIO

Official Technical Roadmaps & Architecture Guides

AI & DEV TOOLS

Cursor AI: Composer & Shadow Workspace Guide

How modern AI coding agents perform multi-file repository indexing, AST dependency mapping, and zero-head branch shadow mutations.

1. AST Codebase Indexing

Parses entire project files into Abstract Syntax Trees locally. Maps function call hierarchies, type signatures, and cross-file import graphs.

```
$ git diff --stat origin/main
14 files modified, 482 insertions(+)
```

2. Background Shadow Worktree

Never edits working branch directly. Spawns an isolated git shadow worktree, applies edits, and runs compiler checks before showing changes.

```
git worktree add -d /tmp/.cursor-shadow
HEAD
npm run typecheck --silent
```

3. Spec-Driven Generation

Breaks large refactors into discrete steps: Plan ➔ Diff ➔ Compile Test ➔ Rollback on Error ➔ Human Approval.

- Zero hallucinated syntax
- Guaranteed compilation

4. Developer Best Practices

Always keep `.cursorrules` in repo root. Pin strict linters and test commands for autonomous agent self-correction.

```
# .cursorrules
strict_types = true
test_command = 'pytest -q'
```